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# Comment on "Mean force potential for the calcium–chloride ion pair in water" [J. Chem. Phys. 99, 4229 (1993)]

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The interionic potential of mean force (pmf) for the $Ca^{++}$–$Cl^{-}$ ion pair in water is computed using the molecular dynamics computer simulation technique. The calculated pmf indicates a stable contact pair (CIP) and a solvent-separated pair (SSIP) centered at 2.9 and 5.0 Å with a 2.8 kcal/mol barrier to dissociation. The SSIP well is about 2.0 kcal/mol deeper than the CIP suggesting that water molecules in the first hydration shell are strongly coordinated to the $Ca^{++}$ ion. Our results do not agree with the pmf reported recently by Guardia, Robinson, Padro [J. Chem. Phys. **99**, 4229 (1993)]. Possible reasons for the discrepancy are discussed. 

In a recent paper, Guardia, Robinson, and Padro examine the detailed mean force and the corresponding potential of mean force (pmf) for the calcium–chloride ion pair in water.[1] Their pmf indicates no minimum at short interionic distance and they attribute this effect to the strength of the $Ca^{++}$–water interactions which prevent the formation of the contact ion-pair configuration. This is an interesting finding. However, as will be shown in this article, with a careful choice of potential parameters and computational methodology, it is possible to obtain a pmf for the calcium–chloride ion pair in water that exhibits well defined contact and solvent-separated wells.

The intermolecular potential functions for water–water, water–ion, and ion–ion interactions consist of coulombic and Lennard-Jones terms. The SPC/E water model is used to describe the water–water interactions.[2] The $Ca^{++}$ ion potential parameters are derived to reproduce the experimental measurements. The $Cl^{-}$ ion potential parameters are derived to match the nonadditive model, which has been refined to fit many properties of experimental measurements.[3] Ion–ion cross terms are obtained using the Lorentz–Berthelot rules.[4] The potential parameters for water, calcium, and chloride ions are presented in Table I. Table II summarizes the thermodynamic and structural properties for a solvated $Ca^{++}$ ion in water. The pmf is calculated using the free energy perturbation method[5] and the molecular dynamics technique. The system consists of 213 water molecules plus one $Ca^{++}$ ion and two $Cl^{-}$ ions in a periodic cube with lengths of 18.656 Å. The reaction coordinate is defined as the distance between the $Ca^{++}$ ion and one $Cl^{-}$ ion and is maintained constant at each ion–ion separation by zeroing the forces and velocities of the ions. The other $Cl^{-}$ ion is allowed to move freely. A prime reason for including an extra $Cl^{-}$ ion is to neutralize the net charge of the system. The long-range interactions are evaluated using the Ewald summation technique.[6] A total of 15 simulations are performed; a typical simulation consists of a 10 ps equilibration period followed by 80 ps of data collection for each $Ca^{++}$–$Cl^{-}$ separation with a timestep of 2 fs. The SHAKE procedure[7] is adopted to constrain all the bond lengths to their equilibrium values and a constant temperature algorithm[7] is used to keep the temperature at 300 K.

The resultant pmf and the primitive model are shown in Fig. 1. The numerical value of the pmf at 6.0 Å was chosen to match the corresponding primitive model value. Upon examination of Fig. 1, we find that the pmf indicates both a stable contact pair (CIP) and a stable solvent-separated pair (SSIP) centered at 2.9 and 5.0 Å with a 2.8 kcal/mol barrier to dissociation. The SSIP well is about 2.0 kcal/mol deeper than the CIP suggesting that water molecules in the first hydration shell are strongly coordinated to the $Ca^{++}$ ion. Figure 1 also displays the computed pmf's as a function of the accumulation time. In all cases, the shapes of the pmf's are the same, and the simulation with 80 ps of averaging for each separation appears to converge well. During the dynamics simulation, we monitored the distance between the free $Cl^{-}$ ion and the complex $Ca^{++}$–$Cl^{-}$ ion pair and found the counter ion is far enough from the complex to be relatively unimportant. Figure 2 illustrates the whole trajectory of the ion–ion distance as a function of the simulation time. The average distance between the free $Cl^{-}$ ion and the complex $Ca^{++}$–$Cl^{-}$ ion pair is about 8.9 Å for the entire trajectory.

Our computed pmf is significantly different from that reported by Guardia *et al.* We use a completely different set of ion–water and ion–ion interaction potentials. Our ion po-

TABLE I. Potential parameters for models of waters and ions: Lennard-Jones parameters ($\sigma$ and $\epsilon$), charges $q$.

| Atom type | $\sigma$(Å) | $\epsilon$(kcal/mol) | $q$ |
|---|---|---|---|
| OW[a] | 3.169 | 0.1553 | −0.8476 |
| HW[a] | 0.000 | 0.0000 | +0.4238 |
| $Ca^{++}$ | 2.895 | 0.1000 | +2.0000 |
| $Cl^{-}$ | 4.401[b] | 0.1000[b] | −1.0000[b] |

[a]The water parameters are taken from Ref. 2.
[b]The ion parameters are taken from Ref. 3.

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TABLE II. Structure and thermodynamics properties of the $Ca^{++}$ ion in water at 300 K.

| | MD | Expt. |
|---|---|---|
| $R_{CaO}$ (Å) | 2.50 | 2.42[a] |
| $R_{CaH}$ (Å) | 3.10 | |
| Coordination number | 8.0 | 5.5-10.0[b] |
| $\Delta H_{sol}$ (kcal/mol) | −385 | −399[c] |

[a]Reference 8.
[b]Reference 9.
[c]Reference 10.

tential parameters are refined to reproduce the experimental enthalpy and structural properties. Our system is net charge neutral and is appropriate for the Ewald summation method. Regardless of the above reasons, our study indicates that with a careful choice of interaction potential parameters and computational methodology, a pmf for the $Ca^{++}$–$Cl^{-}$ ion pair in water that shows well defined stable contact and solvent-separated wells is obtainable. We suggest that Guardia *et al.* should re-evaluate their work before making general conclusions on these ionic systems.

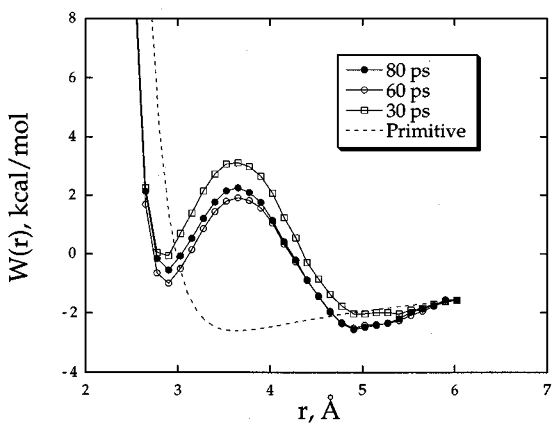


FIG. 1. The pmf for the association of the $Ca^{++}$–$Cl^{-}$ ion pair in water near 300 K as a function of the simulation time. The primitive model (dashed lines) was obtained by dividing the Coulombic term of the total potential by the dielectric constant of the SPC/E water model.

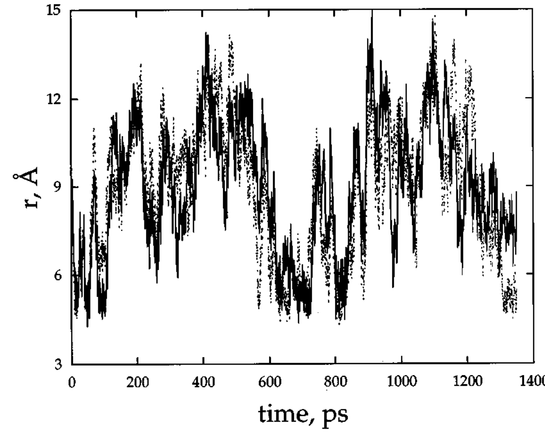


FIG. 2. The distance between the free $Cl^{-}$ ion and $Ca^{++}$ ion (solid), and $Cl^{-}$ ion (dashed), respectively as a function of simulation time.

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